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Exploring ChatGPT's Role in Higher Education: Perspectives from Pakistani University Students on Academic Integrity and Ethical Challenges

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Abstract: This study investigates the perceptions of university students in Pakistan on the use of ChatGPT in learning. It aims to understand the students' views on the benefits, challenges, and overall impact of ChatGPT on their academic experiences. A quantitative approach was employed, collecting data from 400 students across various universities in Pakistan through a structured questionnaire. The validity and reliability of the questionnaire were ensured through pilot testing and statistical analysis. The study found that while most students perceive ChatGPT as a beneficial tool for improving their learning, significant concerns remain about academic integrity, over-reliance on artificial intelligence (AI), and the accuracy of information provided by ChatGPT. The findings suggest that perceptions vary across demographic factors such as age, gender, and type of university. These findings can benefit policymakers and educational institutions in effectively integrating AI tools such as ChatGPT into educational systems, ensuring that the value is maximized while addressing potential disadvantages. The study contributes to the existing body of knowledge on AI utilization in education, particularly within the context of developing countries.

Keywords: ChatGPT; large language models; artificial intelligence in education; educational technology; higher education; ethical consideration in AI



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1. Introduction

The development of artificial intelligence (AI) has significantly impacted various sectors, including education. ChatGPT, developed by OpenAI, is among the AI tools significantly impacting educational systems. This tool has gained widespread recognition for its ability to generate human-like text, accelerating various educational applications. In Pakistan, the integration of ChatGPT into educational frameworks is an increasing trend, stimulating both interest and concern among students and educators.

The findings from existing studies on using AI tools like ChatGPT in educational settings reveal that these technologies offer significant benefits. For example, ChatGPT can provide personalized feedback, assist with writing, and enhance the comprehension of

complex topics. However, concerns about academic integrity, over-reliance on technology, and the potential negative effect on critical thinking skills have been raised. These mixed perceptions highlight the need for an in-depth examination of student attitudes toward ChatGPT within Pakistan's unique educational context.

Although Pakistan has made considerable technological advancements, the education sector faces both opportunities and challenges with the introduction of AI tools like ChatGPT. It promises to bridge educational gaps by providing accessible support. Yet, it also raises questions regarding the preparedness of students and educators to effectively utilize such advanced technologies while upholding academic standards and integrity.

The significance of studying student perceptions of ChatGPT in education lies in its potential to transform learning experiences. AI technologies offer the promise of personalized, efficient, and accessible education. Understanding how students perceive these tools is crucial for maximizing their benefits and addressing potential drawbacks. Educational resources are often limited, but leveraging AI could significantly enhance learning outcomes and bridge educational disparities. This topic is important because it sheds light on the readiness of students to adopt new technologies, their perceived advantages and challenges, and the broader implications for educational policy and practice. By examining these perceptions, stakeholders can make informed decisions on integrating AI into educational systems effectively, ensuring that it supports rather than hinders student learning.

Extensive research has been conducted to explore the applications of ChatGPT in various contexts, including its reliability and validity in educational settings. These studies have tested ChatGPT with diverse sample sizes and user groups, using different statistical methods and comparing ChatGPT with other models. The Technology Acceptance Model (TAM) is often utilized to understand the acceptance and usage of new technologies like ChatGPT. According to TAM, perceived usefulness and ease of use are crucial factors influencing users' acceptance of technology.

This paper utilizes a quantitative approach to explore student perceptions, covering a large population of students from Pakistan's public and private universities. By analyzing the data, the study aims to uncover the detailed views of students on the impact of ChatGPT on their learning processes, academic performance, and overall educational experience. The findings are intended to guide policymakers and educational institutions in harnessing the potential of AI to enhance education while addressing the challenges it presents. The study focuses on the following research questions:

1. How do university students' perceptions of ChatGPT's benefits vary across different demographic groups (e.g., age, gender, type of university)?
2. How do university students in Pakistan perceive the effectiveness of ChatGPT in enhancing their academic performance?

Based on the aforementioned research questions, the following hypotheses are proposed:

1. There is a significant difference in the perceived benefits of ChatGPT on student learning across different age groups, with older students (25–31 and 32–38) perceiving more benefits than younger students (18–24).
2. There is no significant difference in the perceived benefits of ChatGPT between male and female students.
3. Private university students perceive more benefits from using ChatGPT in their learning than government and semi-government university students.
4. Students' primary concerns about using ChatGPT in education (e.g., accuracy, academic integrity, dependency on AI) do not significantly differ according to gender.

In conclusion, this research contributes to the growing literature on AI in education by focusing specifically on the Pakistani context. It aims to provide a balanced view of

the use of ChatGPT, acknowledging its transformative potential for educational practices and the importance of ethical implementation. As Pakistan navigates the integration of AI in education, understanding student perceptions will be crucial to ensuring such technologies' effective and responsible use. The findings can also guide future educators in effectively incorporating AI interactive technologies into their teaching strategies, helping them prepare students for a technologically integrated learning environment while addressing ethical considerations.

2. Literature Review

2.1. Students' Perception

[Shafqat and Amjad \(2024\)](#), explored university students' perceptions of the role of ChatGPT in enhancing their academic performance and their potential concerns about the ethical and social implications of using the tool for completing their academic tasks. The study was conducted on students from private universities in Lahore, Pakistan. Most respondents indicated that ChatGPT is an effective tool that offers practical support in generating innovative ideas for solving academic tasks. However, the respondents indicated concern about the lack of a clear policy on using ChatGPT to write their academic tasks. [Farhi et al. \(2023\)](#) conducted a study on 388 students from two universities in Alain City in the UAE. They found that students considered ChatGPT a revolutionary technology that can assist students in many ways. They also found that ChatGPT usage significantly affects perceived ethics among students. [Bodani et al. \(2023\)](#) surveyed 525 participants from those residing in Karachi, Pakistan; 76.2% of the participants were familiar with ChatGPT, with 51.4% indicating that they did not use the tool frequently, 50.1% indicating that using the AI tool could reduce their cognitive skills, and 40% indicating that they have no concern about encouraging privacy and security issues while using the AI tool. Most participants also indicated that they had moderate trust in the information provided by ChatGPT.

[Singh et al. \(2023\)](#) surveyed 430 students in an MSc computer science program at the University of Hertfordshire, UK. These findings suggest that students are familiar with the AI tool but do not use it regularly for academic purposes. The students strongly urged the university to provide clearer guidelines on using the tool for learning activities. [Imran and Lashari \(2023\)](#) conducted a study on a population of 24 randomly selected students from two private universities in Karachi. Most students thought ChatGPT reduces creative writing, while the rest thought it could be beneficial under controlled conditions. [Mogavi et al. \(2023\)](#) explored social media for answers to the user experience and views of adopting ChatGPT across different educational sectors. They found that it is most commonly used in K-12 education, higher education, and practical skills training. [Albayati \(2024\)](#) developed an integrated model consisting of the technology acceptance model and four constructs, security, privacy, trust, and social inclusion, to study the potential benefits and challenges of integrating ChatGPT into education. [Tossell et al. \(2024\)](#) surveyed the students before and after a writing assignment in an engineering course using ChatGPT. The students reported finding that ChatGPT is a valuable learning tool; however, they were uneasy with its ethical use. In addition, they reported that ChatGPT had poor accuracy, and students had limited feedback on the confidence of its output. [Yilmaz et al. \(2023\)](#) conducted a study on 239 students in science and mathematics education programs at a private university in Almaty, Kazakhstan. They reported an overall positive perception of ChatGPT, with some differences between male and female perceptions regarding ease of use.

[Ngo \(2023\)](#) surveyed 230 students on their perception of using ChatGPT in learning. In general, the students had a favorable perception, reporting saving time, information in various areas, and personalized tutoring and feedback among the benefits of the tool. However, students expressed concern about the inability to assess the reliability and quality

of sources and the inability to cite sources accurately. [Sila et al. \(2023\)](#) analyzed the perception of 107 foundation program students at a private university. The student expressed concern regarding the accuracy and reliability of information provided by ChatGPT but acknowledged that they use ChatGPT to improve their learning experience without impacting their performance. [Obenza et al. \(2024\)](#) explored the perception of 500 university students in the Philippines. Their results show that the students showed a high level of knowledge and understanding of both the advantages and disadvantages of using ChatGPT in higher education.

2.2. Teachers' Perception

[Shakil and Siddiq \(2024\)](#) surveyed 30 ESL teachers from different public sector universities in Pakistan to explore the perception of the use of ChatGPT as detrimental to the analytic writing skills of graduate students. Results revealed the challenge of using ChatGPT in ESL classrooms because students misused the tool for producing general writing with ChatGPT and thus plagiarized the subject matter. They concluded that it results in the decline of the analytical writing skills of ESL students. [Kanwal et al. \(2023\)](#) explored the inclusion of ChatGPT in Pakistani higher education from teachers' perspectives. They surveyed 600 university teachers. Results indicated that ChatGPT can improve learning experiences by offering quick access to information and accommodating varied student demands. However, challenges of overreliance, ethical consequences, and reducing critical thinking were also revealed. [Iqbal et al. \(2022\)](#) used the Technology Acceptance Model to study the attitude of 20 Pakistani university faculty members toward using ChatGPT. Results indicated that they have a negative perception of using this tool. They raised concerns about plagiarism and cheating. They also noted benefits such as the ease of lesson planning and assessment.

[Firat \(2023\)](#) studied both the perceptions of teachers and students. For this study, 7 teachers and 14 Ph.D. students were surveyed. Some themes that emerged from the findings include the "evolution of learning and education systems", "impact on assessment and evaluation", and the "changing roles of educators". [Espartinez \(2024\)](#) conducted a study with 27 participants (15 students and 12 professors) using a survey, interviews, and the ranking of 36 statements on using ChatGPT. They found that the perceptions varied regarding ChatGPT integration and thus concluded with recommendations for optimizing the use of ChatGPT in higher education. [Cambra-Fierro et al. \(2024\)](#) found that ChatGPT adoption increases faculty happiness and energy and reduces stress levels.

2.3. ChatGPT's Role in Teaching and Learning

[Iftikhar et al. \(2024\)](#) explored the perception of learning in higher education institutions in Pakistan regarding using ChatGPT to enhance learning English as a second language. The learners indicated that they find ChatGPT effective in their learning. [Japoshvili-Ghvinashvili and Suleman \(2023\)](#) explored the use of ChatGPT in designing activities for language learners and content-based tasks. The study reported the effectiveness of using ChatGPT and indicated some of the challenges and implications. [Faiz et al. \(2023\)](#) explore and propose a balance between using ChatGPT to aid writing learning. They explore effective strategies for using the tool to improve vocabulary, grammar, and composition while allowing the student to craft logically structured and convenient written discourse. [Rafique et al. \(2024\)](#) conducted a linguistic analysis of the impact of ChatGPT on language evolution. They studied the capabilities of ChatGPT in things like changes in wording, syntax, and semantic elements. Language learning is not the only area; ChatGPT is used in teaching and learning. For example, [Ali et al. \(2024\)](#) explored using ChatGPT to generate multiple-choice questions (MCQs) for a medical and surgical course and compare them

to those written by faculty members. The study reported no significant difference in the quality of questions; however, the faculty members outperformed ChatGPT by developing more grammatically correct questions.

Saif et al. (2024) conducted a study on 156 students from universities in Pakistan in their final year studying management sciences and found that students under stress are more susceptible to using ChatGPT to help them achieve their tasks. They found that, after use, the stress level was reduced. Shah et al. (2024) targeted both students and teachers from universities in Sindh, Pakistan, and used a conceptual model for predictors that influence the acceptance of ChatGPT usage in academia. Abbas et al. (2023) investigated the outcomes of using ChatGPT for university-level instruction and research. Moreover, 200 university faculty and students were chosen using convenience sampling. The majority of the respondents indicate that the integration of ChatGPT had a positive impact on the quality of research and education outcomes. Shehri et al. (2023) investigated the effect of ChatGPT on student performance across universities in Pakistan. The questionnaire was distributed to 37 students from different universities. The results show a positive correlation between using ChatGPT and academic performance. Amjad et al. (2024) investigated the effect of social media, ChatGPT, and M-learning on students' academic performance. The study was conducted on 600 students, and the study concluded that M-learning had a significant impact on student academic performance. However, social media and ChatGPT require strategic integration to enhance performance.

Raza (2023) studied the potential and barriers to using ChatGPT in the social sciences, humanities, pharmacy, and business education at a private university in Pakistan. They present the possibilities and challenges of integrating the AI tool into these disciplines. Shihab et al. (2023) presented their view of using ChatGPT in business education for customer service in particular, as well as the potential of ChatGPT in education and language learning. They also compare ChatGPT with other AI tools. Woithe and Filipec (2023) analyzed the factors that influence the adoption of ChatGPT and its effect on the learning experience in higher education. Rawas (2024) examined the drawbacks and benefits of adopting ChatGPT in higher education. The author concluded that ChatGPT represents a significant opportunity for institutions to improve their accessibility and quality of education; however, he warns that the implementation must proceed with caution and clear understanding.

2.4. ChatGPT's Impact on Research

Arshad et al. (2024) surveyed 223 researchers from Pakistan, where 90% indicated that they are familiar with the use of ChatGPT in research, while 86% indicated that they use the basic version of ChatGPT in their research and 14% use the paid version; 46% indicated their satisfaction with ChatGPT in research. Osama et al. (2023) indicated that ChatGPT is used in scientific research to complement the writing skills of researchers in Pakistan and other countries where researchers are not proficient in English. They do, however, stress that ChatGPT cannot replace human expertise, but it can help in writing skills and communicating the findings of research studies. Rane et al. (2023) explored the use of ChatGPT in research and indicated why ChatGPT cannot be an author and why researchers should be very careful in using ChatGPT, as it could lead to unethical practices such as plagiarism. Aithal and Aithal (2023) explored the use of ChatGPT in research of various types: exploratory, empirical, and experimental. The authors concluded that ChatGPT is being utilized in research and that it impacts scientific publications.

2.5. Ethical Considerations of Using ChatGPT

Schlagwein and Willcocks (2023) discussed in detail the ethics of using AI tools such as ChatGPT in research and science and concluded that, currently, there is no policy for the use of these tools in research but that there is a thin line where the use of these tools can be deemed unethical. Guleria et al. (2023) used experiments to test the usage of ChatGPT in research, especially in medicine and engineering. They found that the information provided by the tool was inaccurate and could have serious implications in these fields. They urge the scientific community to develop policies on using ChatGPT in research. The ethics of using ChatGPT is also explored in many studies (Hualpa, 2023; Mhlanga, 2023; Vargas-Murillo et al., 2023; Stahl & Eke, 2024; Stepanechko & Kozub, 2023; Bang & Park, 2023).

3. Materials and Methods

3.1. Research Design

This study employs a quantitative research design to investigate university students' perceptions of ChatGPT as a supportive educational tool. The research aims to gather data on various aspects of ChatGPT usage, including perceived benefits, challenges, and overall impact on students' learning experiences. This design allows for the collection of quantifiable data that can be statistically analyzed to understand trends and relationships.

3.2. Participants

A total of 400 students from various universities across four provinces of Pakistan—Punjab, Sindh, Khyber Pakhtunkhwa (KPK), and Baluchistan—participated in this study. Participants were selected using convenience sampling to ensure diverse representation across different demographics. The inclusion criteria required that students be enrolled in a university program and have used ChatGPT at least once in their academic activities. The demographic breakdown of the participants is as follows:

- Gender: The sample consisted of 284 females (71%) and 116 males (29%).
- Age Groups: Participants were categorized into four age groups: 18–24 years (264 students, 66%), 25–31 years (70 students, 17.5%), 32–38 years (33 students, 8.25%), and 39 years and above (33 students, 8.25%).
- Type of University: Among participants, 60% were from government universities, 30% from private universities, and 10% from semi-government universities.
- Year of study: Participants were enrolled in various degree programs, including Diploma (6 students, 1.5%), Bachelor of Studies (BS) (251 students, 62.75%), Master of Studies (MS) (98 students, 24.5%), and Doctor of Philosophy (PhD) (45 students, 11.25%).

3.3. Instrument

Data were collected using a structured questionnaire to capture students' perceptions of ChatGPT. The questionnaire was divided into four main sections:

- Section 1: Demographic Information: This section collected basic demographic information, including age, gender, academic level, and the type of institution (public, private, or semi-government).
- Section 2: Student Insights on ChatGPT Use: This section consisted of 12 items measuring students' perceptions of ChatGPT. This section included items on whether students had used ChatGPT, the frequency of use, the version of ChatGPT used, the purposes for using ChatGPT, and the languages used.
- Section 3: Perceived Benefits of ChatGPT on Student Learning: This section included ten items focusing on students' perceptions of using ChatGPT in their education. Items also addressed perceptions of ChatGPT's benefits across various aspects, such

as fostering understanding of course materials, effectiveness, time-saving, support for multiple languages, and personalized tutoring.

- Section 4: Student Concerns and Ethical Perspectives on ChatGPT: This section comprised 14 items exploring concerns about the accuracy of information, academic integrity, dependency on AI, privacy and data security, and the reliability of ChatGPT's responses.

The questionnaire items were developed based on a literature review and validated through expert reviews and a pilot study to ensure clarity and reliability. For the validation of the questionnaire, expert opinions were sought to ensure the validity of all items. Expert feedback led to adjustments in the wording and structure of specific questions to improve comprehensibility. The reliability of the entire instrument was assessed using Cronbach's Alpha. The variables in the study were operationalized on the basis of a review of the related literature. First, the variables were defined conceptually, and measurable sub-factors were identified. These sub-factors included usage and perceptions, perceived benefits, and concerns and ethical considerations. Statements were formulated for each sub-factor, using measurable indicators derived from the literature. Additionally, professional bilingual professors translated the questionnaire into Urdu to ensure accuracy and comprehension for all participants.

3.4. Data Collection

Data collection was conducted using Google Forms, a widely accessible and user-friendly platform for administering online surveys. This method allowed for efficient data collection and immediate digital storage, facilitating subsequent analysis. The Google Forms were distributed to participants through WhatsApp and email after obtaining formal consent from their universities.

3.5. Data Analysis

The collected data were analyzed using both descriptive and inferential statistical methods. Descriptive statistics, including means, standard deviations, and frequency distributions, were calculated to summarize the participants' responses. These statistics provided an overview of the general trends in students' perceptions of ChatGPT.

Inferential statistics were employed to examine relationships and differences between groups based on gender, age, and year of study/degree program. Chi-square tests were used to assess associations between categorical variables, while *t*-tests and ANOVA were conducted to identify differences in mean scores across different groups. All analyses were performed using the trial version of the Statistical Package for the Social Sciences (IBM SPSS V20).

3.6. Ethical Considerations

The study was conducted in accordance with the ethical guidelines of the involved institutions. Participants were informed about the voluntary nature of their participation and their right to withdraw from the study at any time without any consequences. Confidentiality and anonymity of the participants were strictly maintained throughout the research process. Data were stored securely and only accessible to the research team.

3.7. Limitations

While the study provides valuable insights into the perceptions of ChatGPT among university students in Pakistan, certain limitations must be acknowledged. First, reliance on convenience sampling may limit the generalizability of findings to the broader student population. Second, data collection through online questionnaires might have excluded

students with limited digital access, introducing a potential selection bias. Third, self-reported responses are subject to social desirability and recall biases, potentially affecting the accuracy of the data. Lastly, the analysis using the trial version of SPSS may have constrained advanced statistical evaluations.

3.8. Instrument Details

The questionnaire consists of various items, excluding the demographics section. Table 1 details the items in each section:

Table 1. Details of questionnaire sections.

Section 2: Student Insights on ChatGPT Use	
7. Have you used ChatGPT for educational purposes?	8. How frequently do you use ChatGPT for your studies?
9. Which version of ChatGPT do you primarily use?	10. For what purposes do you use ChatGPT in your education? (Select all that apply)
11. In which language(s) do you use ChatGPT for educational purposes?	12. If you use ChatGPT in both English and Urdu, have you noticed a difference in performance between the two languages?
13. Using ChatGPT is easy.	14. ChatGPT provides answers quickly.
15. Using ChatGPT makes me less likely to think deeply.	16. ChatGPT functions like a search engine.
17. ChatGPT supports many input languages.	18. ChatGPT is useful for studying.
Section 3: Perceived Benefits of ChatGPT on Student Learning	
19. Do you believe using ChatGPT fosters a deeper understanding of your course materials compared to traditional study methods?	20. How would you rate the overall effectiveness of ChatGPT in assisting your educational needs?
21. The paid version of ChatGPT offers more accuracy and usefulness than the free version.	22. ChatGPT helps save study time.
23. ChatGPT offers information across various fields.	24. ChatGPT can be used to translate study materials.
25. ChatGPT helps to understand theories and concepts.	26. ChatGPT boosts writing efficiency and productivity.
27. ChatGPT provides personalized tutoring and feedback.	28. ChatGPT enhances learning with personalized experiences.
Section 4: Student Concerns and Ethical Perspectives on ChatGPT	
29. What are your primary concerns about using ChatGPT in education? (Select all that apply)	30. How do you assess the reliability of information provided by ChatGPT for your academic work?
31. Do you have concerns about academic integrity when using ChatGPT for assignments or projects?	32. Do you believe ChatGPT has the potential to replace traditional learning tools and methods?
33. How do you perceive the impact of ChatGPT on the future of education?	34. Would you recommend the integration of ChatGPT into formal education curriculums?
35. How could the use of ChatGPT be improved in your education?	36. ChatGPT sometimes gives unreliable information.
37. ChatGPT may produce false factual references.	38. ChatGPT struggles with citing sources accurately.
39. ChatGPT has difficulty with word replacements and idioms.	40. ChatGPT's response quality drops in longer interactions.
41. ChatGPT can show logical errors and contradictions.	42. ChatGPT struggles to provide an understanding of complex mathematical formulas.

4. Results

In this section, the findings of the study are summarized, highlighting key trends and differences in students' perceptions of ChatGPT. The analysis focuses on demographic variables, including age, gender, and type of university, to explore how these factors influence students' views on the use of ChatGPT in their academic work.

4.1. Results for University Students' Perceptions of ChatGPT (Section 3)

This section presents the result of the data analysis of perceived benefits of ChatGPT on student learning on the basis of demographic variables, including age, gender, and type of university.

An independent samples *t*-test was conducted to compare the perceived benefits of ChatGPT in student learning scores between males and females (see Table 2). There was no significant difference in the scores for the perceived benefits of ChatGPT on student learning between males ($M = 3.67$, $SD = 0.63$) and females ($M = 3.74$, $SD = 0.49$); $t(174.65) = 0.97$, $p = 0.332$ as shown in Table 2.

Table 2. Perceived benefits of ChatGPT on student learning (Section 3) on a gender basis.

	Gender	N	m	SD	df	t-Value (2-Tailed)	Sig.
Perceived Benefits of ChatGPT	F	284	3.7366	0.48925	398	0.97	0.332
	M	116	3.6733	0.62837			

Table 3 comprised results of one-way analysis of variance (ANOVA) that was conducted to evaluate the relationship between age group and the perceived benefits of ChatGPT in student learning (Table 2). Calculated values for the perceived benefits of ChatGPT on student learning indicate a significant statistical difference ($p < 0.05$, $F = 3.962$) in the age mean scores of university students. The mean for each age group is as follows: 18–24 ($M = 3.67$, $SD = 0.504$), 25–31 ($M = 3.86$, $SD = 0.51$), 32–38 ($M = 3.86$, $SD = 0.61$), and 39 or above ($M = 3.56$, $SD = 0.64$). The total sample mean is 3.71 ($SD = 0.533$). It was found that the age group 32–38 has the highest mean score compared to other age groups in terms of perceived benefits of ChatGPT in students' learning.

Table 3. Perceived benefits of ChatGPT on student learning (Section 3) regarding their age groups.

Factor	Age	N	M	SD	df	F	Sig.
Perceived Benefits of ChatGPT on Student Learning	18–24	264	3.67	0.504	3	3.962	0.008
	25–31	70	3.86	0.510	396		
	32–38	33	3.87	0.610	399		
	39 or above	33	3.56	0.647			
	Total	400	3.71	0.533			

According to the result of the statistical analysis presented in Table 4, the mean score for perceived benefits of ChatGPT on student learning has a significant statistical difference ($p < 0.5$, $F = 4.32$) based on the type of institution. The Government group ($n = 314$) had a mean score of 3.69 ($SD = 0.543$), the Private group ($n = 46$) had a mean of 3.92 ($SD = 0.453$), and the Semi-Government group ($n = 40$) had a mean of 3.63 ($SD = 0.485$). The total sample ($N = 400$) had a mean of 3.71 ($SD = 0.533$). It was found that private universities have the highest mean score compared to other universities regarding the perceived benefits of ChatGPT on student learning.

Table 4. Perceived benefits of ChatGPT on student learning (Section 3) regarding their type of university.

Factor	Institution Type	N	M	SD	df	F	Sig.
Perceived Benefit of ChatGPT on Student Learning	Government	314	3.69	0.543	2	4.328	0.014
	Private	46	3.92	0.453	397		
	Semi-Government	40	3.63	0.485	399		
	Total	400	3.71	0.533			

Table 5 shows the result of a one-way ANOVA that was conducted to explore the impact of students' level/degree on the perceived benefits of ChatGPT on student learning. According to table values, the mean score for the perceived benefits of ChatGPT on student learning has a significant statistical difference ($p < 0.5$, $F = 7.80$) based on the student's level/degree. Descriptive statistics were as follows: Diploma ($M = 3.78$, $SD = 0.479$), BS ($M = 3.66$, $SD = 0.491$), MS ($M = 3.92$, $SD = 0.511$), and PhD ($M = 3.53$, $SD = 0.676$). The total sample ($N = 400$) had a mean score of 3.71 ($SD = 0.533$). Analysis revealed that the mean score for the MS group ($M = 3.93$, $SD = 0.51$) was significantly higher than the BS group ($M = 3.67$, $SD = 0.49$) and the PhD group ($M = 3.54$, $SD = 0.68$).

Table 5. Perceived benefits of ChatGPT on student learning (Section 3) regarding their study program.

Factor	Student's Level/Degree	N	M	SD	df	F	Sig.
Perceived Benefits of ChatGPT on Student Learning	Diploma	6	3.78	0.479	3	7.800	0.000
	BS	251	3.66	0.491	396		
	MS	98	3.92	0.511	399		
	PhD	45	3.53	0.676			
	Total	400	3.71	0.533			

4.2. Results for University Students' Perceptions of ChatGPT (Section 2) Based on Gender

A chi-square test of independence was performed to examine the association between gender and the use of ChatGPT for educational purposes. The result was statistically significant, $\chi^2(1, N = 400) = 5.885$, $p = 0.015$. Females were more likely than males to report using ChatGPT for educational purposes. There was no significant difference in the frequency of using different ChatGPT versions for studies, $\chi^2(4, N = 400) = 6.186$, $p = 0.186$, based on gender. The cost may be a factor in students' choice of version, with most opting for freely available options. The students view ChatGPT as a tool for acquiring and processing information rather than for more specific tasks like writing assistance or language learning. English remains the dominant language for academic interactions with AI, even in a multilingual context. Both females (48.9%) and males (46.6%) perceived that ChatGPT performs better in English. This perception could influence students' language choices when using the tool. The high level of perceived ease of use could be a factor in students' widespread adoption of ChatGPT. Most female (83.8%) and male (77.6%) students agreed or strongly agreed that ChatGPT provides answers quickly. This perceived speed could be a critical factor in students' decision to use ChatGPT as an educational tool.

Interestingly, a considerable proportion of both female (59.2%) and male (54.3%) students agreed or strongly agreed with this statement. This suggests that while students find ChatGPT useful, they are also aware of potential drawbacks in terms of cognitive engagement. Female (77.1%) and male (69.0%) students agreed or strongly agreed with this statement. This perception could influence how students use ChatGPT, potentially treating it as an alternative or supplement to traditional search engines. Females were more

likely than males to agree or strongly agree that ChatGPT supports many input languages (59.9% vs. 59.5%). However, a larger proportion of males (15.5%) than females (7.0%) disagreed or strongly disagreed with this statement. This gender difference in perception could influence how students of different genders approach using ChatGPT in multilingual contexts. There was a significant gender difference in perceived usefulness, which explains the higher adoption rate of ChatGPT for educational purposes among female students.

4.3. Results for University Students' Perceptions of ChatGPT (Section 2) Based on Their Age

A chi-square test of independence was performed to examine the relation between age and the use of ChatGPT for educational purposes. The relation between these variables was significant, $\chi^2(3, N = 400) = 17.05, p < 0.001$. The majority of respondents across all age groups reported using ChatGPT for educational purposes, with the highest usage (95.5%) among the 18–24 age group. The frequency of ChatGPT use for studies was not significantly associated with age, $\chi^2(12, N = 400) = 17.29, p = 0.139$. Across all age groups, the most common frequency of use was “Occasionally” or “Rarely”, with fewer respondents reporting “Always” or “Never” using ChatGPT for their studies. The vast majority of users across all age groups (92.25%) reported using the free version of ChatGPT. Across all age groups, research and information gathering were the most common purposes, followed by use as a study aid. English was the predominant language used across all age groups, with some users reporting use in both English and Urdu. Across all age groups, the majority of respondents who used both languages reported that English performed better than Urdu.

The majority of respondents across all age groups agreed or strongly agreed that ChatGPT was easy to use, with the highest agreement (94.7%) in the 18–24 age group. The majority of respondents across all age groups agreed or strongly agreed that ChatGPT provides answers quickly. However, a notable proportion of respondents across all age groups showed a common concern about the potential impact on critical thinking. The majority of respondents across all age groups agreed or strongly agreed that ChatGPT functions like a search engine, with the highest agreement (78%) in the 18–24 age group. Across all age groups, the majority of respondents agreed or strongly agreed that ChatGPT supports many input languages, indicating a general awareness of ChatGPT’s multilingual capabilities. A high proportion of respondents across all age groups agreed or strongly agreed that ChatGPT is useful for studying, with the highest agreement (81.8%) in the 18–24 age group.

4.4. Results for University Students' Perceptions of ChatGPT (Section 2) Based on Type of Institution

A chi-square test of independence was performed to examine the relationship between institution type and the use of ChatGPT for educational purposes. The relationship between these variables was insignificant, $\chi^2(2, N = 400) = 0.966, p = 0.617$. The majority of students across all institution types reported using ChatGPT for educational purposes.

Across institution types, occasional and rare use were the most common frequencies reported. The free version was overwhelmingly the most used across all institution types.

Research and information gathering was the most common purpose across institution types, followed by use as a study aid. English was the predominant language used across all institution types. Across all institution types, most students reported that English performs better.

The majority of students across all institution types agreed or strongly agreed that using ChatGPT is easy, with government university students showing the highest proportion of agreement. Most students across all institution types agreed or strongly agreed that ChatGPT provides answers quickly. A considerable number of students across all institution types agreed that using ChatGPT makes them less likely to think deeply. The

majority of students across all institution types agreed or strongly agreed that ChatGPT functions like a search engine. Most students across all institution types agreed that ChatGPT supports multiple input languages. An overwhelming number of students across all institution types agreed or strongly agreed that ChatGPT is useful for studying.

4.5. Chi-Square Results for University Students' Perceptions of ChatGPT (Section 2) Based on Study Program

A chi-square test of independence was performed to examine the relationship between the students' level/degree and the use of ChatGPT for educational purposes. The relation between these variables was significant, $\chi^2(3, N = 400) = 15.33, p = 0.002$. Students in BS programs reported the highest rate of ChatGPT use (94.8%), followed by MS (89.8%), PhD (77.8%), and diploma programs (83.3%). PhD students reported more frequent use, with 42.2% using it "Always" or "Frequently" compared to 19.9% of BS and 24.5% of MS students. The free version was most common across all programs (92.3% overall). Research and information gathering was the most common purpose across programs (35% overall).

Use of English only was highest among PhD students (86.7%) compared to other programs. Across all programs, the majority of the students (48.25%) reported that English performs better. Compared to other programs, MS students had the highest rate of agreement (81.6% "Agree" or "Strongly Agree"). A large majority across all programs (82% overall) agreed or strongly agreed that ChatGPT provides answers quickly. However, a notable proportion across all programs (57.75% overall) agreed or strongly agreed that ChatGPT reduces deep thinking. BS and MS students were more likely to agree that ChatGPT functions like a search engine (75.3% and 85.7%, respectively) than PhD students (53.3%). MS students showed the highest agreement (69.4% "Agree" or "Strongly Agree"), with the statement that ChatGPT's support for multiple input languages, while PhD students showed the lowest (37.8%). A large majority across all programs (83% overall) agreed or strongly agreed that ChatGPT is useful for studying.

4.6. Results for University Students' Perceptions of ChatGPT (Section 4) Based on Gender

A chi-square independence test was applied to check the primary concerns about the accuracy of information, academic integrity, dependency on AI for learning, privacy, and data security of university students while using ChatGPT based on their gender. An insignificant association ($X^2 = 27.96, df = 24, p = 0.262$) was found between male and female students regarding their concerns about using ChatGPT in education.

Regarding the question of whether university students assess the reliability of the information provided by ChatGPT for their academic work based on their gender, the response options ranged from "always verify with additional sources" to "do not verify". An insignificant association ($X^2 = 6.81, df = 3, p = 0.078$) showed that both genders responded similarly regarding information verification from other sources. A significant association ($X^2 = 6.671, df = 2, p = 0.036$) was found between male and female university students regarding their concerns about academic integrity when using ChatGPT for assignments. An insignificant relation ($X^2 = 3.25, df = 2, p = 0.197$) was found between male and female university students regarding their beliefs that ChatGPT has the potential to replace traditional learning tools and methods. Male and female students reported similarly about the impact of ChatGPT on the future of education: positively, it is a beneficial tool, and negatively, it poses more challenges than benefits. Some male and female students recommended the integration of ChatGPT into formal education curricula, whereas some did not, and some recommended it with certain restrictions.

The chi-square independence test was applied to see how the use of ChatGPT could be improved in the education of university students based on gender. A non-significant relation ($X^2 = 5.26, df = 7, p = 0.628$) was found between male and female university students

regarding the use of ChatGPT in improving education. Many male and female university students agree that ChatGPT sometimes gives unreliable information, produces false factual references, and struggles with citing sources accurately. Both genders were of the opinion that ChatGPT has difficulty with word replacements and idioms, and ChatGPT's response quality drops in longer interactions. Many students opined that ChatGPT can show logical errors and contradictions. Similarly, both genders reported that ChatGPT struggles to provide an understanding of complex mathematical formulas.

4.7. Results for University Students' Perceptions of ChatGPT (Section 4) Based on Age

A chi-square independence test was applied to check the primary concerns of university students regarding using ChatGPT in education based on their age. The chi-square results ($\chi^2 = 102.96$, $df = 72$, $p = 0.010$) were found significant at the 5% significance level for university students while assessing their primary concerns regarding using ChatGPT in education. This suggests that age does not significantly influence how participants verify information from ChatGPT. The concerns about academic integrity when using ChatGPT do not significantly differ across age groups. Age does not significantly affect participants' views on ChatGPT's potential to replace traditional learning methods. The perceptions of the impact of ChatGPT on education do not significantly vary across different age groups. Age does not significantly influence participants' recommendations regarding the integration of ChatGPT into education. The results also indicated that suggestions for improving ChatGPT in education do not significantly differ across age groups. Moreover, perceptions of ChatGPT's reliability do not significantly differ across age groups. No significant difference was found in students' concerns about ChatGPT producing false references across age groups. Perceptions of ChatGPT's ability to cite sources accurately do not significantly differ across age groups. No significant difference was found in students' responses regarding difficulty with word replacements and idioms with ChatGPT, and students across all ages showed their agreeableness that ChatGPT's response quality dropped in longer interactions. Students across all age groups are more inclined to agree that ChatGPT can show logical errors and contradictions, and it struggles to provide an understanding of complex mathematical formulas.

4.8. Results for University Students' Perceptions of ChatGPT (Section 4) Based on Type of University

A chi-square test of independence was performed to examine the relationship between institution type and primary concerns about using ChatGPT in education. The relationship between these variables was not significant, $\chi^2(48, N = 400) = 50.39$, $p = 0.379$. Across all institution types, accuracy of information was the most common primary concern ($n = 138$), followed by dependency on AI for learning ($n = 70$) and academic integrity concerns ($n = 47$). The most common approach used by students across all types of institutions was to always verify with additional sources ($n = 148$), followed closely by sometimes verifying ($n = 141$). Only a small portion of respondents ($n = 40$) reported not verifying ChatGPT information. Whereas the majority of respondents across institution types expressed concerns about academic integrity ($n = 251$), while 84 were unsure, and 65 had no concerns. For the belief that ChatGPT has the potential to replace traditional learning tools and methods, the most common response across institution types was "Maybe/Partially" ($n = 159$), followed by "Yes" ($n = 165$) and "No" ($n = 76$). For the impact of ChatGPT on the future of education, the majority of respondents across institution types had mixed feelings about the impact ($n = 217$), while 142 viewed it positively and 41 viewed it negatively. To recommend the integration of ChatGPT into formal education curriculums, the most common response across institution types was "Yes" ($n = 179$), followed by "Maybe, with certain restrictions" ($n = 161$), and "No" ($n = 60$). Across all institution types, providing

more detailed explanations for complex subjects was the most common suggestion ($n = 115$), followed by increasing the accuracy and reliability of responses ($n = 100$) and enhancing integration with educational platforms and tools ($n = 55$). The majority of respondents across institution types either agreed ($n = 184$) or were neutral ($n = 115$) about ChatGPT sometimes giving unreliable information. Only a small portion strongly disagreed ($n = 16$) or strongly agreed ($n = 49$) with this statement and reported that ChatGPT may produce false factual references. Most respondents across institution types were either neutral ($n = 159$) or agreed ($n = 156$) that ChatGPT has difficulties with citing sources accurately and ChatGPT has difficulty with word replacements and idioms. Most respondents were either neutral ($n = 163$) or agreed ($n = 137$) that ChatGPT's performance may decrease in extended conversations. In contrast, a majority of respondents either agreed ($n = 158$) or were neutral ($n = 145$) about this potential limitation. Similarly, most respondents were either neutral ($n = 162$) or agreed ($n = 149$) that ChatGPT may have difficulties in providing an understanding of complex mathematical formulas.

4.9. Results for University Students' Perceptions of ChatGPT (Section 4) Based on Study Program

A chi-square test of independence was performed to examine the relationship between the students' level/degree program and primary concerns about using ChatGPT in education. The relation between these variables was not statistically significant, $\chi^2(72, N = 400) = 88.26, p = 0.120$. Across all students' levels/degrees, accuracy of information was the most common primary concern ($n = 138$), followed by dependency on AI for learning ($n = 70$) and academic integrity ($n = 47$). The most common approach across all programs was to sometimes verify with additional sources ($n = 141$), followed closely by always verifying ($n = 148$). Similarly, a majority of respondents ($n = 251$) expressed concerns about academic integrity. The most common response across programs was "Maybe/Partially" ($n = 159$).

When they were asked about their belief that ChatGPT has the potential to replace traditional learning tools and methods, most respondents had mixed feelings about the impact ($n = 217$) of ChatGPT on the future of education across all institutions. Regarding recommending the integration of ChatGPT into formal education curriculum, the most common response was "Maybe, with certain restrictions" ($n = 161$). Across all degree programs, providing more detailed explanations for complex subjects was the most frequent suggestion ($n = 115$), followed by increasing the accuracy and reliability of responses ($n = 100$). However, the distribution of preferences varied somewhat according to the degree programs, contributing to the significant result. The majority of respondents across all programs either agreed ($n = 184$) or were neutral ($n = 115$) about this limitation. In contrast, most respondents were either neutral ($n = 165$) or agreed ($n = 125$) that ChatGPT may produce false factual references. A large proportion of respondents either agreed ($n = 156$) or were neutral ($n = 159$) regarding this limitation of ChatGPT with citing sources accurately, and ChatGPT has difficulty with word replacements and idioms.

The majority of respondents were either neutral ($n = 163$) or agreed ($n = 137$) that ChatGPT's response quality drops in longer interactions. Most respondents either agreed ($n = 158$) or were neutral ($n = 145$) regarding this limitation of ChatGPT to show logical errors and contradictions. For ChatGPT's struggle to understand complex mathematical formulas, the most common responses were neutral ($n = 162$) or agree ($n = 149$) across all programs.

The analysis was based on a combination of the demographics and sections of the questionnaire. Each section of the questionnaire was analyzed based on the demographics of age, gender, type of university, and specialization. Thus, the findings revealed notable differences in perceptions across these demographic groups. For instance, older students

demonstrated a higher mean score than other age groups in terms of the perceived benefits of ChatGPT in students' learning. Detailed results related to demographics are explained in their respective portions of the paper. These demographic-based results emphasize the need for tailored strategies when introducing and integrating AI technologies into educational contexts, ensuring equitable benefits across diverse groups.

5. Discussion

The findings from this study offer significant insights into Pakistani university students' perceptions of ChatGPT as a tool in their academic activities. The results indicate that students generally perceive ChatGPT as beneficial for their learning, with specific demographic factors influencing these perceptions.

The study shows that age significantly affects how students perceive the benefits of ChatGPT. Older students, particularly those in the 25–31 and 32–38 age groups, reported higher perceived benefits than their younger counterparts in the 18–24 age group. This could result from increased maturity and potentially due to the demanding academic responsibilities faced by older students, which might make them more appreciative of the services offered by ChatGPT. However, students aged 39 and above showed slightly lower perceived benefits, suggesting a reluctance to fully embrace new technology or a preference for more traditional study methods. The 39 and older age group is nowhere near the digital native generation and includes those who are used to less reliance on technology and more reliance on traditional resources such as books, libraries, notes, etc.

The type of university students attend also plays a crucial role in shaping their perceptions of ChatGPT. Students from private universities reported significantly higher perceived benefits compared to those from government or semi-government institutions. This difference might reflect resource variations, technology exposure, or the curriculum's integration of advanced tools like ChatGPT. Often equipped with better technological infrastructure, private institutions may allow students to explore and utilize such tools effectively.

Gender, however, did not significantly influence the perceived benefits of ChatGPT, with both male and female students showing similar levels of appreciation for the tool. This suggests that the perceived benefit of ChatGPT is widely spread across the student body, indicating widespread acceptance.

The study also explored the ethical considerations associated with using ChatGPT. While a significant portion of students recognized the ease of use and the quick responses provided by the tool, concerns were raised about the potential impact on deep thinking and academic integrity. Interestingly, a higher proportion of female students expressed concerns about academic integrity, highlighting a gender difference in how the ethical implications of AI tools are perceived. This aligns with the significant correlation between gender and concerns about ChatGPT, potentially replacing traditional learning methods.

Moreover, the study found that while students acknowledge the multilingual capabilities of ChatGPT, English remains the dominant language for interacting with the tool. This preference suggests that despite the availability of other languages, the perceived effectiveness of ChatGPT in English might be higher, particularly among students who are more comfortable with academic content in English.

The perception of university students is important for teacher education because some of them will be future teachers. Thus, the findings suggest that teacher education programs must prepare future educators to effectively integrate AI-driven interactive tools like ChatGPT into classrooms. By addressing both the benefits and challenges of these technologies, educators can enhance learning while maintaining academic integrity.

6. Conclusions

This study comprehensively explains university students' perceptions of ChatGPT in Pakistan, highlighting varying acceptance and concerns across different demographic groups. The findings underscore the importance of age and institutional type in shaping students' views on the utility of AI tools like ChatGPT in education.

The analysis revealed that most respondents across all demographics perceived ChatGPT as a valuable tool for enhancing learning. Students widely regarded ChatGPT as easy to use, with many emphasizing its ability to quickly provide answers, which was identified as a key factor driving its adoption as an educational tool. Across all age groups and institution types, the primary use of ChatGPT was for research and information gathering, followed by its use as a study aid. English was the predominant language used, with respondents noting better English performance than Urdu. Additionally, students expressed general awareness of ChatGPT's multilingual capabilities and appreciated its flexibility in supporting various input languages. However, there were noticeable differences in perceptions based on demographics. Female students were more likely than male students to report using ChatGPT for educational purposes and perceived it as more useful. Across all groups, students valued ChatGPT's ability to function like a search engine, simplifying the process of accessing information. However, concerns were raised regarding ChatGPT's potential to reduce critical thinking, reflecting a shared apprehension about over-reliance on the tool.

Common issues among the participants included the accuracy of information, dependency on AI for learning, and concerns about academic integrity. Participants consistently noted challenges such as ChatGPT's occasional delivery of false references, logical errors, and difficulty understanding complex mathematical formulas. While these limitations were acknowledged, most respondents relieved their concerns by verifying ChatGPT's output with external sources, a practice reported consistently across all groups. Perceptions about ChatGPT's potential to replace traditional learning tools and methods were mixed across all demographics, with many respondents suggesting conditional support for its integration into formal education. Suggestions for improvement included providing more detailed explanations for complex subjects, increasing the accuracy and reliability of responses, and enhancing integration with existing educational platforms.

Respondents also highlighted that ChatGPT's performance tends to decrease in longer interactions and acknowledged its struggles with producing reliable citations. Despite these challenges, most participants across all groups viewed ChatGPT positively, recognizing its usefulness for research, information gathering, and study support. This overall perception underscores ChatGPT's potential as an educational tool while identifying improvement areas.

While the tool is generally perceived as beneficial, notable concerns exist about its impact on critical thinking and academic integrity, particularly among female students.

The results suggest that for ChatGPT to be effectively integrated into educational frameworks, especially in diverse contexts like Pakistan, there needs to be a balanced approach that maximizes its benefits. ChatGPT has the potential to revolutionize education by providing instant, personalized feedback and facilitating language learning. Educational institutions should consider these demographic differences when promoting the use of AI tools and provide clear guidelines to ensure their ethical use.

Further research should explore the long-term impacts of ChatGPT on learning outcomes. There is a potential for ChatGPT to significantly enhance learning outcomes by providing personalized learning experiences and instant feedback. Investigating strategies to mitigate any negative effects on students' cognitive engagement is crucial. Additionally, enhancing the tool's effectiveness in languages other than English could broaden its accessibility and utility in multilingual educational settings.

The findings suggest that teacher education programs can be enhanced by incorporating training on AI-driven interactive technologies like ChatGPT. This preparation will enable future educators to effectively integrate these tools into their teaching practices, ensuring a balanced approach that maximizes benefits while addressing potential challenges. This study contributes to the growing discourse on AI in education by offering insights specific to the Pakistani context, which can inform policy and practice as educational systems evolve in the digital age.

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